Budgeted Page-Level Preprocessing Selection for Vietnamese Administrative OCR: A Controlled Comparison of Learned, Retrieval-Based, and Static Shortlists

Cong Thanh Bui[1†], Anh Nguyen Van[1*†], Hung Nguyen The[1], Tu Nguyen Anh[1], Quynh Le Thi[1], Linh Lai Thi[1], Dung Nguyen Trung[1]

[1]National Laboratory for Securing Information (NLSI), Vietnam Ministry of Defence, Hanoi, Vietnam.

*Corresponding author(s). E-mail(s): nguyenanhbh14@gmail.com;
Contributing authors: congthanhttmt@gmail.com; hungnguyenthe1211@gmail.com; nguyenanhtu789@gmail.com; lequynh.k51.ktqs@gmail.com; linhlai1605@gmail.com; ntdtoanud2011@gmail.com;
†These authors contributed equally to this work and share first authorship.

Abstract

Purpose: OCR on Vietnamese administrative documents is sensitive to page artifacts and preprocessing choices. We study budgeted selection of page-level preprocessing candidates from a fixed pool. **Methods:** A stratified five-parameter design yields 50 preprocessing configurations. We train recognizer-specific CER regressors on OCR-free page features and configuration parameters. We compare learned surrogate, nearest-neighbor transfer (NN-K), FixedTopK, and random shortlists with nested GroupKFold on 300 Vietnamese administrative first pages in raw and contour-suppressed settings, and on 626 SROIE receipts. A nested oracle-free selector chooses one output from a surrogate top-3. **Results:** Surrogate and NN-K shortlists significantly outperform random shortlists in all evaluated combinations of recognizer, setting, and budget. FixedTopK remains highly competitive on the Vietnamese data, while the surrogate significantly outperforms it in all eight SROIE comparisons. Across all 32 configurations, the gap from FixedTopK to the sampled-pool oracle is positively associated with each policy's advantage over FixedTopK for Surrogate, Random, and NN-K; Random's correlation is larger in 30/32. This headroom is therefore a policy-agnostic retrospective measure, not evidence specific to learned ranking. The selector has lower mean CER than FixedBest in all eight settings; six pairwise two-sided comparisons have $p < 0.05$. **Conclusion:** No shortlist policy is uniformly best. Controlled comparisons show the value of page-conditioned selection over random sampling, and the nested selector recovers part of the shortlist gain without reference text.

Keywords: document OCR, preprocessing selection, algorithm selection, surrogate-based ranking, nearest-neighbor transfer, oracle-free output selection

1 Introduction

Optical character recognition (OCR) on administrative documents remains sensitive to seemingly low-level image preparation choices. In Vietnamese administrative documents, two issues are

especially prominent. First, stamps, decorative headers, and logos can cause a detector to generate false or low-quality text crops. Second, the effectiveness of denoising, contrast enhancement, brightness correction, and sharpening varies substantially across pages, making any single static preprocessing recipe unreliable.

The straightforward solution to the second issue is exhaustive per-image search over preprocessing combinations, but this is expensive because every candidate requires a full OCR pass. This creates a practical ranking problem: can a model trained once on previously observed documents identify a small shortlist of promising preprocessing configurations for a new page?

This paper studies that question in a page-level OCR pipeline built around PaddleOCR detection and three different recognizers. The proposed framework treats preprocessing choice as budgeted page-conditioned shortlist selection over a fixed candidate pool. A stratified design over a five-parameter preprocessing space is used to construct a fixed 50-candidate pool; each candidate is then evaluated with OCR to obtain observed CER values. Page-level features are extracted once from each page, and learned surrogate, retrieval-based, and static shortlist policies are compared under identical fixed-budget evaluation.

Two questions determine whether this formulation is practically useful. First, how do learned surrogate ranking, retrieval from visually similar pages, and a static shortlist compare under the same OCR-call budget? Second, can a promising shortlist be reduced to one output without using reference text at inference time? We address the first question through nested, page-grouped comparisons against Random K and FixedTopK, and the second through a nested oracle-free top-3 selector. We also analyze the per-page FixedTopK-to-oracle headroom as a retrospective property of the candidate pool. Because headroom and any FixedTopK-relative policy advantage share the FixedTopK term, we include Random and NN-K controls and do not interpret their correlation as evidence specific to learned ranking.

The paper makes the following contributions:

1. It formulates page-level OCR preprocessing as a budgeted shortlist-selection problem over a fixed candidate pool and evaluates it across recognizers, preprocessing conditions, and a second document domain.
2. It provides a controlled nested GroupKFold comparison of cross-instance surrogate ranking, nearest-neighbor transfer, static FixedTopK, and random shortlist policies under identical OCR-call budgets.
3. It evaluates a nested oracle-free top-3 output selector that selects one OCR result from a surrogate shortlist without reference text at inference time.
4. It audits FixedTopK-to-sampled-oracle headroom with Surrogate, Random, and NN-K controls, separating a policy-agnostic retrospective measure of available improvement from evidence about which policy exploits that improvement.

This article is related to an earlier conference paper by the authors on hybrid OCR for Vietnamese administrative documents [2]. The earlier work focused on an end-to-end OCR pipeline that combined PaddleOCR detection, VietOCR recognition, stamp handling, and language-model-based post-processing. In contrast, the present article addresses a different methodological question: how page-level preprocessing candidates can be prioritized under a fixed OCR budget. Compared with the conference paper, the present study introduces a fixed candidate-pool formulation, recognizer-specific surrogate modeling, static and retrieval-based shortlist baselines, nested GroupKFold evaluation, paired statistical comparisons, oracle-free top-3 output selection, and evaluation on the SROIE receipt dataset.

The paper is organized as follows. Section 2 reviews related work. Section 3 describes the datasets and task definition. Section 4 formalizes the proposed method. Section 5 details the experimental setup. Section 6 reports the main results. Section 7 discusses interpretation and practical implications. Section 8 outlines limitations, and Section 9 concludes.

2 Related Work

2.1 OCR Systems and Document Preprocessing

Modern modular OCR pipelines, including PaddleOCR/PP-OCR, separate detection from

recognition, with detected text regions cropped before recognition [4, 10]. Our pipeline follows this structure with PaddleOCR detection and three alternative recognizers based on different design assumptions: Tesseract [21], EasyOCR [8], and the Vietnamese-specific VietOCR [17]. Vietnamese document analysis and OCR is an active application area facing complex diacritics, limited annotated data, and real-world document variation [9], with representative work spanning scene-text recognition and the VinText benchmark [13], administrative-document digitization [14], and MC-OCR receipt recognition [28]. Against this background, this paper studies adaptive page-level preprocessing selection as an orthogonal axis to recognizer choice; the recognizers themselves remain unmodified.

Document image preprocessing has a long history in OCR. Global thresholding such as Otsu's method [15] and local/adaptive methods such as Sauvola and Pietikäinen [19] are canonical baselines, with DIBCO-style evaluations established as benchmarks for historical and degraded documents [18]. Beyond binarization, OCR pipelines often use photometric enhancement and denoising operations such as contrast-limited adaptive histogram equalization (CLAHE) [29], bilateral filtering [25], gamma correction, and unsharp masking; these directly motivate the parameter space evaluated here. Recent learned restoration methods, including DE-GAN [23] and DocEnTr [24], learn image-to-image enhancement transforms and are complementary to our setting, which instead learns a cross-instance surrogate that ranks lightweight preprocessing parameter settings by downstream OCR CER. Administrative documents also contain artifacts such as stamps, decorative headers, and watermarks; this motivates the contour-suppression setting evaluated here, while the main contribution remains budgeted prioritization of preprocessing candidates rather than artifact removal.

2.2 Algorithm Selection, Meta-Learning, and Surrogate Optimization

Surrogate and sequential model-based optimization methods are also relevant to our setting. Methods such as SMAC [7], Bayesian optimization frameworks [22], and TPE-based sequential optimization [1] guide a sequential optimization run: configurations are evaluated iteratively, and the search model or distribution is updated from observations collected during that run. In standard formulations, this process is optimized for the current task rather than reused unchanged to rank a fixed pool on unseen document pages. In contrast, our surrogate is trained once across many document pages and reused at inference to rank preprocessing candidates for a new page without per-image OCR feedback.

Our data collection strategy is also closer to design-of-experiments than to iterative Bayesian optimization. Space-filling designs such as Latin hypercube sampling [12] are useful for broadly covering the preprocessing parameter space during offline data collection.

Closer to adaptive preprocessing, Garg and Chaudhury [5] studied automatic selection of document-image enhancement parameters using image-quality assessment and expected recognition accuracy within an EM-based optimization framework. Vats et al. [27] applied Bayesian optimization to hyperparameter selection in document image binarization, and Singh et al. [20] learned surrogate models for document image quality metrics to support automated document image processing. These studies support the relevance of adaptive parameter selection and surrogate modeling in document preprocessing. However, the studies by Garg and Chaudhury and by Vats et al. focus on parameter optimization and binarization, whereas Singh et al. use cross-image surrogate modeling for document-image quality metric estimation rather than OCR-CER-based shortlist ranking.

2.3 Positioning of This Work

This work sits at the intersection of OCR preprocessing, algorithm selection, and surrogate modeling. Unlike OCR-engine papers, it does not introduce a new detector or recognizer. Unlike classical preprocessing studies, it does not propose one fixed filter or binarization rule. Unlike standard Bayesian optimization, it does not perform iterative per-image search. Compared with prior automatic preprocessing parameter-selection work, our formulation shifts the problem from estimating a single optimal preprocessing setting to selecting a small candidate shortlist under a fixed

OCR budget. This enables direct comparison of static, retrieval-based, and learned shortlist policies, where the key requirement is not exact CER prediction but reliable prioritization of candidate preprocessing settings.

Three empirical findings structure the paper's contributions. First, page-conditioned shortlist construction (surrogate and NN-K) reliably outperforms random shortlist selection in all evaluated comparisons (24/24 Vietnamese and 8/8 SROIE) under identical OCR-call budgets, even with moderate surrogate R^2. FixedTopK is stronger than Random K and remains highly competitive on the Vietnamese data, so the learned, retrieval-based, and static policies form a setting-dependent ordering rather than a single uniformly dominant policy. Second, the apparent association between the gap from FixedTopK to the sampled-pool oracle and each policy's advantage over FixedTopK is policy-agnostic: the same coupled relationship appears for Random and NN-K, while direct policy-to-policy controls do not support an interpretation specific to learned ranking. Third, a nested oracle-free top-3 selector recovers part of the shortlist gain without reference text at inference time, has lower mean CER than FixedBest in all eight settings, and six of the eight paired two-sided comparisons reach $p < 0.05$. To our knowledge, this specific problem formulation—budgeted page-conditioned shortlist ranking for OCR preprocessing, with controlled comparison of learned, retrieval-based, static, and random policies plus nested oracle-free output selection—has not been explicitly studied in the OCR preprocessing literature. This paper evaluates it across Vietnamese administrative documents, contour-suppressed variants, three recognizers, and the SROIE receipt dataset.

3 Dataset and Task Definition

3.1 Vietnamese Administrative Document Dataset

The primary dataset consists of 300 Vietnamese administrative document images obtained from the Vietnamese Government portal (https://chinhphu.vn/he-thong-van-ban) and spanning the years 2015 to 2024, with balanced temporal sampling of 30 images per year. Source documents were treated at the page level: when a source PDF contained multiple pages, only the first page was converted to an image and used in this study. Thus, each sample corresponds to a first-page document image rather than a complete multi-page document. Each page is paired with a manually curated transcription prepared by the authors and used as the reference text for page-level CER computation. The transcriptions were manually checked by the authors against the corresponding page images before CER computation. Before evaluation, OCR outputs and reference transcriptions underwent the same whitespace normalization: all whitespace sequences, including tabs and line breaks, were collapsed to single spaces. No case folding, punctuation removal, or explicit Unicode normalization was applied by the evaluation code. Because the corpus is drawn from a single public government document portal, it captures temporal variation in document appearance and image quality while limiting broader source-domain variation.

3.2 Task Definition

The task is defined as follows: given a document page image, produce OCR text for the full page. Performance is evaluated by page-level character error rate (CER) against the corresponding reference transcription. The optimization target is end-to-end page-level CER: preprocessing parameters are applied to the full page before text detection, so the objective captures effects on both detection quality and recognition accuracy, rather than recognition-only performance on pre-cropped text regions.

3.3 OCR Pipeline

The OCR pipeline used throughout the study consists of optional contour suppression on the full page, page-level preprocessing, PaddleOCR text-region/text-line detection, recognition with one of the three recognizers, and page-level CER computation against the reference transcription. PaddleOCR is used as a fixed shared detection stage across recognizers, allowing the study to isolate the effect of page-level preprocessing selection while holding detection fixed. This ordering is important because preprocessing is applied before the shared detection stage, so it can affect both the text regions found by PaddleOCR and

the image crops passed to the recognizer. Page-level artifacts such as stamps, decorative headers, logos, and watermarks may cause false, missed, or low-quality detections; once text is missed or incorrectly cropped, a downstream recognizer cannot fully recover it. The study therefore optimizes preprocessing at the page level rather than as a recognition-only crop enhancement step.

3.4 SROIE Receipt Dataset

The second dataset is the ICDAR 2019 SROIE benchmark [6], used to evaluate the framework in a second document setting with a different domain and OCR language configuration. To remove duplicate copies in the supplied flat file structure, the collection script excludes image and transcription filenames containing `(` and retains only images with a matching transcription stem. This yields 626 receipt images with matching full-text receipt transcriptions. The SROIE bounding-box annotations are not used for CER computation. SROIE is evaluated with English OCR configurations: PaddleOCR detection with `lang=en`, Tesseract recognition with `eng`, and EasyOCR recognition with `en`. VietOCR is omitted because it is Vietnamese-specific. The SROIE experiment is therefore not intended to reproduce field-level SROIE key-information-extraction evaluation, but to apply the same full-page shortlist-ranking protocol under a different document domain and OCR language configuration. A compact dataset summary is provided in Online Resource 1.

4 Proposed Method

Fig. 1 summarizes the proposed framework, separating offline candidate evaluation and policy construction from inference-time shortlist construction, best-of-K shortlist evaluation, and oracle-free top-3 output selection.

4.1 Problem Formulation

Let $\mathcal{D} = \{(x_i, r_i)\}_{i=1}^{N}$ denote the page images and their reference transcriptions. The primary Vietnamese dataset has $N = 300$, and the SROIE receipt dataset has $N = 626$ after duplicate removal. Let $\Theta = \{\theta_1, \ldots, \theta_M\}$ denote the fixed candidate pool of page-level preprocessing settings, with $M = 50$. For a given experimental setting, define the page representation as $\tilde{x}_i = x_i$ in the raw setting and $\tilde{x}_i = C(x_i)$ in the contour-suppressed setting, where C is the fixed contour-suppression transform described below.

For recognizer e, the observed OCR response for page i and preprocessing candidate θ_j is

$$y_{i,j,e} = \mathrm{CER}\left(O_e(P_{\theta_j}(\tilde{x}_i)), r_i\right), \tag{1}$$

where P_{θ_j} applies the page-level preprocessing candidate and O_e denotes the OCR pipeline with the shared detector and recognizer e. Page-level CER is computed from edit distance:

$$\mathrm{CER}(o, r) = \frac{S(o, r) + D(o, r) + I(o, r)}{|r|}, \tag{2}$$

where S, D, and I are the numbers of substitutions, deletions, and insertions needed to transform OCR output o into reference transcription r.

For each page i and preprocessing candidate θ_j, the surrogate input is

$$z_{i,j} = [f(\tilde{x}_i), \theta_j, g(f(\tilde{x}_i), \theta_j)] \in \mathbb{R}^{22}, \tag{3}$$

where $f(\tilde{x}_i)$ contains 12 OCR-free page features, θ_j contains the 5 preprocessing parameters, and $g(f(\tilde{x}_i), \theta_j)$ contains 5 predefined interaction terms between page properties and preprocessing controls, all defined in the remainder of Section 4; the 22-dimensional input thus comprises these three groups concatenated. A recognizer-specific surrogate h_e is trained as a supervised CER regressor that predicts $\hat{y}_{i,j,e} = h_e(z_{i,j})$. Although trained as a CER regressor, the surrogate is used at inference time as a ranking function. The induced page-conditioned top-K shortlist policy is

$$\pi_K^{h_e}(\tilde{x}_i) = \operatorname*{Kmin}_{\theta_j \in \Theta} h_e(z_{i,j}), \tag{4}$$

where Kmin returns the K preprocessing candidates with the lowest predicted CER values.

The practical objective is therefore not exact CER prediction, but page-conditioned shortlist construction under a fixed OCR budget:

$$\min_{\pi_K} \mathbb{E}_i \left[\min_{\theta_j \in \pi_K(\tilde{x}_i)} y_{i,j,e} \right], \quad \pi_K(\tilde{x}_i) \subset \Theta, \quad |\pi_K(\tilde{x}_i)| = K. \tag{5}$$

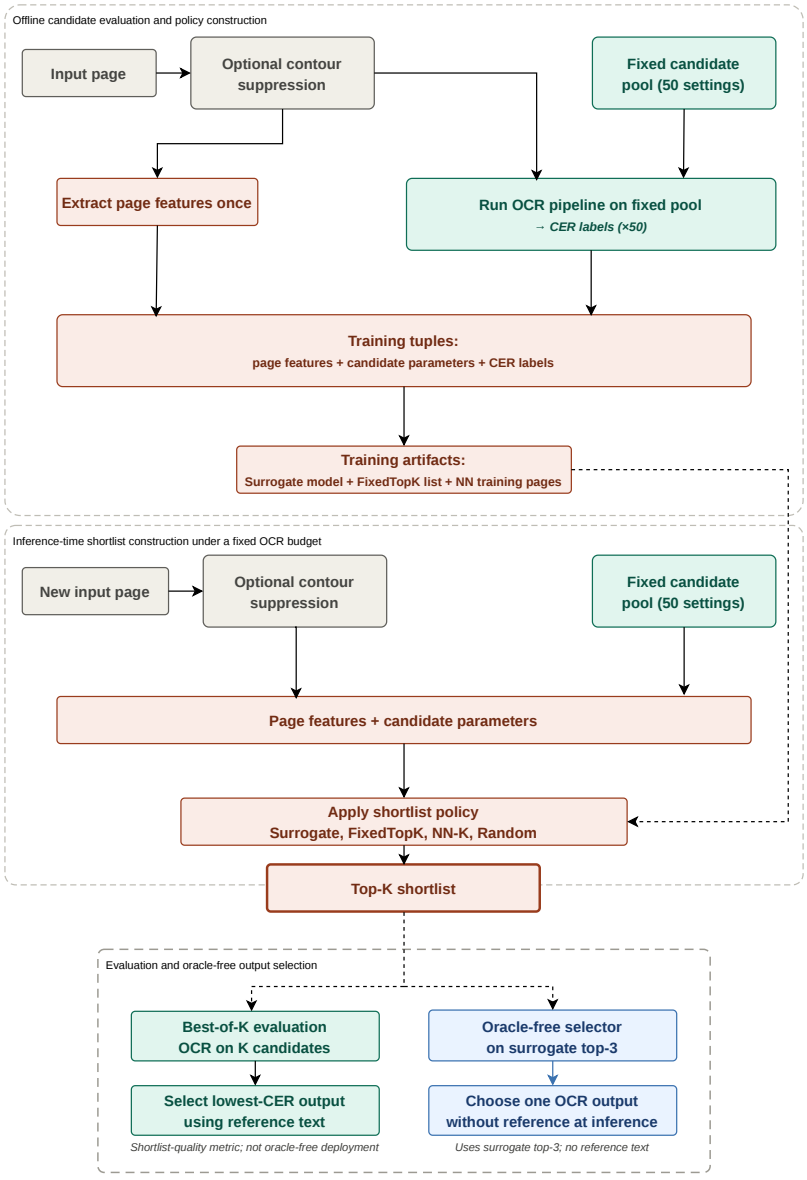


Fig. 1 Proposed page-level preprocessing framework. In the offline stage, each page is optionally contour-suppressed, page features are extracted once, and a fixed pool of 50 preprocessing candidates is evaluated with OCR to obtain observed CER values for recognizer-specific surrogate training and baseline construction. At inference time, shortlist policies rank or transfer candidates for a new page under a limited OCR budget. The lower dashed branches distinguish the best-of-K evaluation protocol used to assess shortlist quality from the oracle-free top-3 selector used as an inference-time deployment check.

This is an induced selection objective rather than the direct training loss. The surrogate is fitted to observed CER values, and its ranking is evaluated by the best observed CER within the selected top-K shortlist.

The reference baselines can be expressed in the same finite candidate pool. The sampled-pool oracle is

$$B_{i,e}^{\mathrm{oracle}} = \min_{\theta_j \in \Theta} y_{i,j,e}, \tag{6}$$

which is the lowest observed CER within the sampled pool rather than a global optimum. The fixed baseline selects the single training-pool candidate with the lowest mean training CER,

$\theta^*_{\text{fixed},e} = \arg\min_{\theta_j \in \Theta} \frac{1}{|\mathcal{I}_{\text{train}}|} \sum_{i \in \mathcal{I}_{\text{train}}} y_{i,j,e}$. Random K samples K candidates uniformly from the same Θ, while NoPre uses the exact OCR run with all preprocessing operations disabled. We also define FixedTopK as a stronger static shortlist baseline: for each outer training fold, the K preprocessing candidates with the lowest mean training CER are selected and then applied as the same K-candidate shortlist to every held-out page.

4.2 Contour Suppression as Upstream Page-Level Preprocessing

As an optional preprocessing step, we use a fixed contour-based rule to identify rectangular regions that meet fixed criteria and replace their pixels with white before text detection. It is intended to reduce large non-text regions that can trigger false detections, such as stamps, decorative headers, logos, and watermarks, rather than to serve as a general artifact-removal method. Given an input page image, the method converts the image to grayscale, applies a Gaussian blur with a 5-by-5 kernel, computes an Otsu threshold, and extracts external contours from the inverted threshold mask. For each contour, a bounding rectangle is computed; if the rectangle satisfies all of the following criteria, it is overwritten with white pixels in the original image: contour area greater than 500 pixels, aspect ratio between 1.0 and 5.0, width greater than 80 pixels, and height greater than 30 pixels. A one-at-a-time robustness check on a 30-image subset found the aspect-ratio upper bound to be stable, while the area, minimum-width, and minimum-height thresholds affected CER more noticeably; we therefore treat contour suppression as a fixed, domain-motivated preprocessing condition rather than a fully optimized artifact-removal module. The output remains a page image rather than a binarized image; only selected large regions are suppressed. The objective is to reduce false detections before OCR, not to directly improve recognition in already detected text crops.

4.3 Page-Level Preprocessing Search Space

The page-level preprocessing search space has five tunable parameters applied to the grayscale page before text detection: `gaussian_ksize` $\in \{0, 3, 5, 7\}$, `bilateral_d` $\in \{0, \ldots, 9\}$, `clahe_clip` $\in [0, 4.0]$, `gamma` $\in [0.5, 1.5]$, and `sharpen` $\in [0, 2.0]$. These parameters correspond to Gaussian blur, bilateral filtering, CLAHE, gamma correction, and sharpening, respectively; their off/identity values and roles are tabulated in Online Resource 1.

These five controls were selected to address common page-level degradations using lightweight operations while keeping the search space compact and compatible with stratified sampling. Gaussian blur and bilateral filtering address noise under different smoothing assumptions, CLAHE targets local contrast and faded text, gamma correction adjusts global brightness, and sharpening addresses blurred strokes. We restrict the search to grayscale intensity operations that preserve page geometry and do not convert the image to a binary representation. Operations such as binarization and morphology would require categorical or pipeline-level design choices outside the fixed five-dimensional search space. Deskewing was not included because skew was minor across the 300 pages as measured by the `orig_skew_angle` feature in the current candidate data: no page exceeded 2°, and geometric correction is outside the intensity-based preprocessing scope studied here.

The two denoising parameters are discrete: `gaussian_ksize=0` disables Gaussian blur, while positive values specify the Gaussian kernel size. Positive Gaussian kernel sizes are restricted to odd values, as required by the OpenCV implementation, and a kernel size of 1 is excluded because it is effectively an identity operation and redundant with the off value. Similarly, `bilateral_d=0` disables bilateral filtering, while positive values specify the bilateral filter diameter. The remaining parameters are continuous, with `clahe_clip=0` disabling CLAHE, `gamma=1.0` serving as the identity brightness transform, and `sharpen=0` disabling sharpening.

These ranges were retained after boundary checks that varied one parameter at a time on a subset of images across the three recognizers in both raw and contour-suppressed settings. The checks supported the Gaussian upper bound clearly and framed the remaining bounds as practical ranges used consistently across all three recognizers rather than recognizer-specific optima.

Implementation settings outside this five-dimensional space are held fixed across all experiments: OpenCV automatic sigma for Gaussian blur, bilateral color and spatial sigma values of 75, an 8-by-8 CLAHE tile grid, and parametric 3-by-3 sharpening. Operations are applied in the fixed order Gaussian blur, bilateral filtering, CLAHE, gamma correction, and sharpening, with denoising before contrast/brightness adjustment and sharpening last to avoid amplifying earlier-stage noise.

4.4 Data Collection via Stratified Candidate Sampling

A fixed candidate pool of 50 preprocessing combinations is generated from a stratified five-dimensional unit-cube design, following the space-filling motivation of Latin hypercube sampling [12], and then reused for every page. The two discrete parameters are obtained by binning the corresponding design coordinates into the discrete values actually used: `gaussian_ksize` into $\{0, 3, 5, 7\}$ and `bilateral_d` into $\{0, \ldots, 9\}$. The three continuous parameters (`clahe_clip`, `gamma`, and `sharpen`) are scaled linearly from their design coordinates into their stated ranges. This Latin-hypercube-style construction gives stratified one-dimensional coverage for the continuous parameters and near-balanced marginal counts for the discrete levels, while treating the 50 candidates as a compact finite design rather than a full factorial grid. For each page and candidate combination, the page is preprocessed, OCR is run, and page-level CER is computed against the reference transcription. This yields $300 \times 50 = 15{,}000$ rows per recognizer in the primary dataset, one for each page i and candidate θ_j, with each row storing page features, preprocessing parameters, and the observed CER after OCR.

The candidate-pool design aims for broad coverage of the preprocessing response surface rather than per-image convergence to an optimum; with $D = 5$ search dimensions, $M = 50$ follows the common $n \approx 10d$ rule of thumb for initial space-filling computer experiments [11]. The sampled-pool oracle reported in the experiments therefore refers to the best observed configuration within this finite pool, not a global optimum over the continuous preprocessing space. The offline OCR evaluation loop is summarized in Online Resource 1.

4.5 Page-Level Features

For each page, image features that do not require OCR are extracted once before evaluating candidate preprocessing settings. In the raw setting, features are computed from the original page image; in the contour-suppressed setting, they are computed from the contour-suppressed page image. The base feature set contains seven intensity and texture descriptors: blur, contrast, brightness, noise, edge density, entropy, and dynamic range. We also include five document-aware descriptors: text density, skew angle, connected-component count, aspect ratio, and estimated line count. The base features capture low-level image quality, tone, and texture, whereas the document-aware descriptors capture page structure that is not represented by intensity statistics alone.

The surrogate input concatenates these 12 page-level features with the five preprocessing parameters. In addition, we include five predefined interaction terms that pair each preprocessing control with the image property it is intended to address: blur $\times$ `gaussian_ksize`, contrast $\times$ `clahe_clip`, brightness $\times$ `gamma`, noise $\times$ `bilateral_d`, and entropy $\times$ `sharpen`. These interactions encode predefined, domain-motivated assumptions about how the effect of each preprocessing operation depends on page properties, without performing data-driven feature selection over a larger interaction pool. The resulting surrogate input has 22 dimensions: 12 page features, 5 preprocessing parameters, and 5 interaction terms.

We use OCR-free hand-crafted page features rather than learned visual embeddings to keep the ranking model lightweight and independent of any additional pretrained image model at inference time. The feature definitions also allow domain-motivated interaction terms that directly pair image properties with the preprocessing controls they are intended to modulate. The effect of these feature groups is evaluated empirically in the ablation study in Section 6; the full feature-group table is provided in Online Resource 1.

4.6 Surrogate Model

For each recognizer, we train a separate surrogate regressor to predict page-level CER for a given page and preprocessing candidate. The input is the 22-dimensional vector described above, and the target is the actual CER obtained after applying the candidate preprocessing setting and running the OCR pipeline. The surrogate is therefore recognizer-specific but shares the same feature representation and candidate pool across recognizers.

The surrogate is used for ranking rather than for exact CER prediction. For each held-out page, the model scores every candidate preprocessing combination in the fixed candidate pool, and candidates with lower predicted CER are ranked higher. We refer to this as a cross-instance surrogate because the model is trained across many pages and reused for unseen pages, rather than being fitted separately for each page through iterative feedback.

We use regression followed by ranking rather than a direct pairwise or listwise learning-to-rank objective for two practical reasons. First, the observed target is a continuous page-level CER, and calibrated CER predictions are useful beyond candidate ordering: they define Top1Pred, prediction margins, and features for the oracle-free top-3 selector. Second, the main methodological question is whether page-conditioned scoring can enrich a small shortlist under a fixed OCR-call budget, not which tabular ranking loss is optimal. Direct learning-to-rank models are therefore left as an extension rather than treated as a primary contribution.

We consider four regression model families: Ridge regression, Random Forest, XGBoost, and LightGBM. These families cover linear, bagged-tree, and gradient-boosted regressors commonly used for feature-based tabular regression. Model-family selection and evaluation are performed within the nested GroupKFold protocol described in Section 5. We do not include stacking in the primary evaluation because group-aware out-of-fold predictions would be needed to train a stacking meta-learner without mixing candidate rows from the same page.

4.7 Evaluation Protocol

We evaluate shortlist quality under fixed OCR-call budgets $K \in \{3, 5, 10, 15\}$. These values span tight to moderate budgets, covering 6% to 30% of the 50-candidate pool. For each held-out page, the surrogate ranks the fixed 50-candidate pool by predicted CER, and the Surrogate K shortlist consists of the K candidates with lowest predicted CER. The Random K baseline samples K candidates uniformly from the same pool; to reduce Monte Carlo noise, random results are averaged over 20 reproducible trials per page without requiring additional OCR runs, since all trials resample from the already evaluated candidate pool. FixedTopK selects the K candidates with the lowest mean training CER in each outer fold and uses that same static shortlist for every held-out page. Surrogate K, Random K, and FixedTopK are evaluated using best-of-K CER, defined as the minimum actual CER among the K candidates in the shortlist.

As an additional retrieval-based baseline, we evaluate a nearest-neighbor transfer shortlist (NN-K) in the Vietnamese raw, contour-suppressed, and SROIE settings. For each outer fold, page features are standardized using only the training pages. For each held-out page, NN-K finds the nearest training pages in this OCR-free feature space, transfers the oracle-best preprocessing candidate from each nearest training page, removes duplicate candidates while preserving neighbor order, and fills any remaining shortlist slots with fold-specific FixedTopK candidates if needed. NN-K therefore uses only training-fold OCR responses for candidate transfer and is evaluated on held-out pages with the same best-of-K protocol as Surrogate K and FixedTopK.

We also report three reference baselines. No preprocessing uses exact OCR runs with all preprocessing operations disabled. FixedBest selects, within each training fold, the single preprocessing combination with the lowest mean CER on the training pages and applies it to the held-out pages. Sampled Oracle is the minimum actual CER among all 50 sampled candidates for each page; it is the lowest observed CER within the sampled candidate pool, not the true global optimum over all preprocessing parameters.

We report computational cost primarily in OCR-call units because OCR runtime depends on

hardware, recognizer implementation, and page content. The offline data collection stage requires 50 OCR evaluations per page and is used only to build the training and evaluation table. At inference time, OCR-free feature extraction and scoring all 50 tabular candidate rows are small compared with running OCR. NoPre, FixedBest, and Top1Pred require one OCR call per page; the oracle-free top-3 selector requires three OCR calls; Surrogate K, NN-K, and FixedTopK require K OCR calls under the best-of-K shortlist-quality protocol; and the sampled-pool oracle requires all 50 OCR calls. Thus, learned ranking adds feature-extraction and model-scoring overhead, but the dominant deployment cost remains the chosen OCR-call budget.

Because Surrogate K and Random K both use best-of-K selection during evaluation, these results measure shortlist quality under a fixed candidate-evaluation budget rather than fully oracle-free deployment performance. The comparison remains fair because both methods receive the same within-shortlist best-of-K evaluation. We additionally compute a deployable one-call operating point, Top1Pred, defined as the single candidate with the lowest out-of-fold predicted CER. Top1Pred does not use reference text or within-shortlist CER selection; it is reported as a supplementary deployment check rather than as the main shortlist-quality metric.

To test whether a short surrogate shortlist can be used without reference text at inference time, we also evaluate an oracle-free top-3 output selector. For each held-out page, OCR is run on the surrogate-ranked top-3 candidates and the selector chooses one of the three OCR outputs. Candidate-level selector features include the surrogate rank, predicted CER and prediction margins, preprocessing parameters, the number of detected crops, OCR text length and character-composition statistics, simple document-pattern counts such as dates and administrative keywords, and agreement features based on normalized edit distances among the three OCR outputs. We compare five selector families: random-forest and logistic-regression classifiers trained to identify the best candidate within the top-3, and random-forest, gradient-boosting, and ridge regressors trained to predict candidate CER. Selector-family choice is nested inside each outer fold: the family with the lowest inner-fold validation CER on the outer-training pages is refit on all outer-training pages and then used on the held-out pages. The held-out selector therefore uses OCR outputs, surrogate predictions, and preprocessing metadata, but not reference text or held-out CER values. We compare it with Top1Pred, the agreement-only baseline, FixedBest, NoPre, and the best-of-3 oracle. The agreement-only baseline selects the top-3 output with the smallest mean normalized edit distance to the other two outputs.

For a feature-matched sensitivity analysis, we replace the surrogate's predicted CER and prediction-margin features by the within-shortlist ranks 1, 2, and 3. The exact surrogate top-3 candidates, OCR outputs, and all remaining selector features are retained, and the nested selector is rerun. This makes the prediction-related inputs comparable with NN-K and FixedTopK, which provide an ordering but not a continuous CER prediction. The analysis tests whether policy comparisons persist without the surrogate's numerical score and margin information. It is not a pure test of shortlist quality because the selector is refit and continues to use OCR-output features.

5 Experimental Setup

5.1 OCR Recognizers

All primary experiments use the same text-detection stage: PaddleOCR detection with Vietnamese configuration (`lang=vi`). The detected text regions are cropped and passed to one of three recognizers: VietOCR, Tesseract with the Vietnamese language pack (`vie`), and EasyOCR with Vietnamese recognition (`vi`). Thus, the detector and page-level preprocessing pipeline are held fixed while the recognizer changes.

The final candidate-response data were generated with PaddleOCR 3.3.2 on PaddlePaddle GPU 3.3.0, VietOCR 0.3.13 using its `vgg_transformer` configuration, EasyOCR 1.7.2, and Tesseract 5.3.4 through `pytesseract` 0.3.13. The installed Tesseract `eng` and `vie` language packages were version `1:4.1.0-2`. For each page, detector polygons were converted to axis-aligned bounding-rectangle crops and processed in the `dt_polys` order returned by PaddleOCR; no additional reading-order sorting was applied. Nonempty recognized strings were joined with tab separators in that preserved order. VietOCR

output was retained only when its predicted confidence was strictly greater than 0.8. Tesseract was called with its default engine and page-segmentation settings.

PaddleOCR was used as the shared detector because it supports a modular pipeline in which one detector can be paired with multiple recognizers. Keeping detection fixed lets the experiments focus on preprocessing selection rather than detector choice. The recognizers were chosen to cover complementary practical OCR backends: VietOCR as a Vietnamese-specific neural recognizer, Tesseract as a mature open-source OCR engine with LSTM recognition, and EasyOCR as a general-purpose neural OCR toolkit. These backends are used here to test whether preprocessing-selection behavior changes across recognizers, not as the main optimization target. For the SROIE setting described in Section 3, the same detector-recognizer structure is used with English configurations, and VietOCR is omitted because it is Vietnamese-specific.

5.2 Nested GroupKFold Cross-Validation

For the Vietnamese experiments, all reported surrogate results are computed with a nested 5-fold GroupKFold protocol, using the page image as the grouping unit. All 50 candidate preprocessing rows from the same page are kept in the same fold, preventing leakage in which candidates from one page appear in both training and test data. Each outer fold therefore evaluates the selected surrogate on held-out pages that were not used to fit that fold's model. After the five outer folds are completed, the held-out page-level results are combined across folds, giving one out-of-fold result for each of the 300 pages. The same group-aware nested protocol is used for the SROIE setting, with results combined over all 626 receipt images.

Model-family selection is nested inside each outer training fold. For each recognizer and outer fold, the candidate model families are compared using inner GroupKFold on the outer-training pages, and the family with the lowest mean validation MAE is selected. That selected family is then refit on the full outer-training fold and evaluated on the corresponding held-out fold. This nested protocol avoids selecting a model family using information from the outer test pages, following standard guidance on reducing selection-induced optimism in cross-validation [3, 16, 26].

A final model is also selected and refit on all available pages for saved inference artifacts, but this final model is not used to compute the reported cross-validation metrics. A compact protocol table and stepwise nested-evaluation algorithm are provided in Online Resource 1.

5.3 Metrics

CER is the primary OCR-quality metric and is computed at the page level against the reference transcription. Surrogate prediction quality is summarized using MAE and R^2. We report two Spearman rank correlations: global Spearman, computed over all held-out candidate rows within an outer fold, and per-image Spearman, computed as the mean within-page candidate-rank correlation over held-out pages. Per-image Spearman is most directly aligned with the shortlist objective because the surrogate must rank the fixed candidate pool separately for each test page. Best-of-K CER evaluates the actual CER achieved by the best observed candidate within the surrogate-ranked shortlist.

For statistical comparison, Surrogate K is compared with Random K and FixedTopK using two-sided paired Wilcoxon signed-rank tests over page-level best-of-K CER values. The supplementary NN-K baseline is compared with Surrogate K, FixedTopK, and Random K using the same two-sided paired page-level tests. The oracle-free top-3 selector is compared with Top1Pred, the agreement-only baseline, and FixedBest using two-sided paired page-level tests. Tests are run separately for each recognizer, setting, and K value where applicable, and the direction of an effect is reported from the paired mean CER difference. Aggregate CER values are reported as means over the held-out page-level results combined across outer folds. We also compute percentile bootstrap confidence intervals for paired mean differences, with Surrogate K minus the corresponding baseline as the signed difference. Random K is estimated as the mean best-of-K CER over 20 reproducible random draws per page.

For the retrospective headroom analysis, let $F_{i,K}$ be the fold-specific FixedTopK CER, O_i the sampled-pool-oracle CER, and $P_{i,K}$ the CER of policy P. We define headroom $H_{i,K} = F_{i,K} - O_i$

and FixedTopK-relative policy advantage $A^P_{i,K} = F_{i,K} - P_{i,K}$. These quantities share $F_{i,K}$, and $A^P_{i,K} \leq H_{i,K}$ because the sampled-pool oracle is the best candidate in the pool. We therefore compute the coupled correlation for Surrogate, Random, and NN-K, and separately correlate headroom with direct policy-to-policy gains that do not contain FixedTopK: Random minus Surrogate, Random minus NN-K, and NN-K minus Surrogate. This control analysis is descriptive and is not used for model or policy selection.

Because the study reports multiple paired comparisons, p-values are interpreted as descriptive evidence within predefined comparison families rather than as a single global null-hypothesis test. We emphasize repeated directionally consistent patterns, paired mean differences, and bootstrap confidence intervals, and we avoid drawing strong conclusions from isolated marginal tests. The strongest claims in the paper are therefore the repeated surrogate-vs-random shortlist improvement, the SROIE surrogate-vs-FixedTopK pattern, and the setting-dependent nature of the surrogate-vs-NN-K comparison.

5.4 Hyperparameters and Sensitivity

The primary results use the nested GroupKFold protocol described above, with model-family selection performed inside each outer training fold. For implementation-level regressor settings, we do not run a broad hyperparameter search. Instead, we compare four predefined surrogate model families, chosen to cover common linear, bagged-tree, and gradient-boosted regressors, with fixed hyperparameters for each family. The goal is not to exhaustively benchmark surrogate regressors, but to evaluate whether a practical cross-instance surrogate can provide useful shortlist rankings. This keeps the focus on candidate ranking under a fixed OCR budget rather than on maximizing a standalone regressor benchmark.

As a sensitivity check, we also reran the surrogate evaluation with default regressor settings in supplementary holdout experiments for both raw and contour-suppressed images. These reruns preserved the main significance pattern: surrogate shortlists remained significantly better than random shortlists in all 12/12 raw comparisons and all 12/12 contour comparisons. We therefore treat the nested GroupKFold results as the primary evidence and the default-parameter reruns as supplementary evidence that the surrogate-vs-random conclusion is not fragile to the chosen regressor settings.

6 Results

6.1 Preprocessing Response Variation in the Fixed Candidate Pool

PaddleOCR detection settings are held fixed throughout the main multi-recognizer experiments. This isolates the page-level preprocessing-selection problem and keeps the detector shared across the three recognizers. We next quantify the observed preprocessing-response variation within the fixed candidate pool.

The sampled preprocessing pool shows substantial response variation across candidates. Across the 50 sampled preprocessing settings per page, the mean within-image CER coefficient of variation is 55.5%, 67.1%, and 25.1% in the raw setting for VietOCR, Tesseract, and EasyOCR, respectively. In the contour-suppressed setting, the corresponding values are 68.9%, 77.5%, and 29.4%. Table 1 summarizes the per-page improvement of the sampled-pool oracle over exact no-preprocessing OCR. In the raw setting, the sampled-pool oracle improves over NoPre on 90–98% of pages depending on the recognizer, with mean relative CER reductions between 15.3% and 35.4%. In the contour-suppressed setting, the corresponding improvement rate is 90–96%, with mean relative reductions between 16.5% and 38.2%. These within-pool oracle values are not deployable results because they require evaluating all 50 candidates. Instead, they show that the candidate pool often contains substantially better preprocessing choices than NoPre. The practical problem is therefore to find such candidates with far fewer OCR calls, which motivates the budgeted shortlist-ranking approach evaluated below.

6.2 Surrogate Model Quality

The surrogate models achieve moderate absolute CER prediction accuracy but useful rank-correlation signal. Across the six Vietnamese recognizer-setting combinations, nested-CV R^2

Table 1 Sampled-pool oracle improvement over exact NoPre. Relative improvement is computed per page as (NoPre − SampOracle)/NoPre.

Setting	Recognizer	Pages improved	Mean rel.	Median rel.
Raw	VietOCR	287/300 (96%)	25.0%	22.7%
Raw	Tesseract	293/300 (98%)	35.4%	36.1%
Raw	EasyOCR	271/300 (90%)	15.3%	14.1%
Contour	VietOCR	282/300 (94%)	27.2%	25.5%
Contour	Tesseract	287/300 (96%)	38.2%	40.5%
Contour	EasyOCR	270/300 (90%)	16.5%	15.4%

ranges from 0.435 to 0.614, global Spearman from 0.522 to 0.676, and mean per-image Spearman from 0.497 to 0.776; full fold-level model-quality metrics are given in Online Resource 1. Per-image Spearman is most directly aligned with the shortlist objective because each test page requires ranking the same 50 preprocessing candidates for that page. The key interpretation is that absolute prediction metrics and global rank correlation alone do not determine shortlist usefulness, so the next section evaluates ranking usefulness directly through best-of-K shortlist performance.

6.3 Surrogate Shortlists vs Random Shortlists

Under the same K-candidate budget, two-sided paired Wilcoxon tests show lower CER for surrogate shortlists than for random shortlists across all three recognizers, all evaluated K values, and both raw and contour-suppressed settings. The raw setting has 12/12 significant surrogate-vs-random comparisons, the contour-suppressed setting also has 12/12, giving 24/24 significant comparisons overall. Best-of-K curves across recognizers and budgets are visualized in Online Resource 1.

The effect-size pattern is consistent with the budgeted-ranking interpretation: surrogate gains over random are largest at small K and generally decrease as K grows, because random shortlists become more likely to contain a good candidate when the shortlist budget increases. The gains are largest for EasyOCR and smallest for VietOCR, consistent with VietOCR already having the strongest baseline performance. The stronger FixedTopK baseline is more competitive than Random K, however, and is analyzed separately below.

6.4 Comparison with FixedTopK Static Shortlists

Table 3 summarizes the comparison with FixedTopK at the tightest and largest evaluated shortlist budgets. FixedTopK is a strong static baseline and is generally much stronger than random selection, although it is nearly tied with random in the lowest-error contour-suppressed VietOCR cases at larger K. Two-sided paired Wilcoxon tests show significant surrogate improvements over FixedTopK at $p < 0.05$ in 5/12 raw Vietnamese comparisons and 8/8 SROIE comparisons. In the contour-suppressed Vietnamese setting, no comparison significantly favors the surrogate, while 2/12 comparisons significantly favor FixedTopK. Bootstrap intervals for the paired mean difference (Surrogate minus FixedTopK) give a compatible interpretation. In the raw Vietnamese setting, four of 12 intervals favor the surrogate, one favors FixedTopK, and seven overlap zero. In the contour-suppressed setting, no interval favors the surrogate, three favor FixedTopK, and nine overlap zero. On SROIE, all eight intervals favor the surrogate. In mean CER, Tesseract favors FixedTopK at $K = 3$ in both raw and contour-suppressed Vietnamese settings, with differences of +0.19 percentage points and +0.33 percentage points, respectively; the latter has a 95% interval of [+0.16, +0.55]. On SROIE, by contrast, all intervals are negative, for example -1.04 percentage points for Tesseract at $K = 3$ with a 95% interval of [-1.33, -0.76]. These results show that the added value of page-conditioned ranking over a strong static shortlist is dataset- and recognizer-dependent.

Table 2 Vietnamese best-of-K CER results. Values are percentages. Each S/R entry reports Surrogate K / Random K.

Setting	Recognizer	NoPre	FixedBest	SampOracle	K=3 S/R	K=5 S/R	K=10 S/R	K=15 S/R
Raw	VietOCR	4.70	4.63	3.61	4.23 / 4.53	4.10 / 4.25	3.87 / 3.98	3.79 / 3.86
Raw	Tesseract	9.82	9.29	6.85	8.29 / 8.98	7.86 / 8.40	7.46 / 7.77	7.20 / 7.50
Raw	EasyOCR	15.44	15.19	13.19	14.27 / 15.75	13.94 / 15.18	13.57 / 14.41	13.42 / 14.07
Contour	VietOCR	3.15	3.19	2.31	2.89 / 3.08	2.77 / 2.84	2.58 / 2.61	2.50 / 2.51
Contour	Tesseract	7.69	7.28	5.19	6.49 / 6.98	6.15 / 6.50	5.70 / 5.94	5.49 / 5.71
Contour	EasyOCR	13.25	13.09	11.10	12.35 / 13.64	11.97 / 13.06	11.58 / 12.31	11.37 / 11.97

Table 3 Comparison with FixedTopK static shortlists. CER values are percentages. Diff is Surrogate minus FixedTopK, so negative values favor the surrogate.

Setting	Recognizer	Surr K=3	FixedTopK K=3	Diff K=3	Surr K=15	FixedTopK K=15	Diff K=15
Raw	VietOCR	4.23	4.27	-0.04	3.79	3.85	-0.07
Raw	Tesseract	8.29	8.11	0.19	7.20	7.30	-0.10
Raw	EasyOCR	14.27	14.33	-0.05	13.42	13.50	-0.08
Contour	VietOCR	2.89	2.87	0.02	2.50	2.51	-0.02
Contour	Tesseract	6.49	6.16	0.33	5.49	5.46	0.03
Contour	EasyOCR	12.35	12.22	0.13	11.37	11.37	0.00
SROIE	Tesseract	43.23	44.27	-1.04	40.99	41.44	-0.44
SROIE	EasyOCR	39.08	39.22	-0.14	38.17	38.28	-0.11

6.5 Policy-Agnostic FixedTopK-to-Oracle Headroom

The aggregate comparison above shows that FixedTopK leaves different amounts of residual improvement across pages and settings. We quantify this as the per-page headroom $H_{i,K} = F_{i,K} - O_i$. A surrogate-only analysis would compare it with $F_{i,K} - S_{i,K}$, the surrogate's advantage over FixedTopK. However, the two quantities share $F_{i,K}$, and the sampled-pool oracle enforces $F_{i,K} - P_{i,K} \leq H_{i,K}$ for every pool-constrained policy P. A positive correlation can therefore reflect this construction rather than a learned-ranking effect.

We audit the relationship using the same held-out pages for Surrogate, Random, and NN-K. As Table 4 shows, each coupled FixedTopK-relative advantage has a positive significant correlation with headroom in all 32 combinations of recognizer, setting, and budget, and the highest-headroom quartile has a larger mean advantage than the lowest quartile in all 32 configurations for every policy. Random's correlation is larger than the surrogate's in 30/32 configurations. Thus, the original 32/32 pattern is policy-agnostic and cannot establish that high-headroom pages specifically favor learned ranking.

We additionally correlate headroom with direct policy-to-policy gains that do not contain FixedTopK. The correlation between headroom and Random minus Surrogate ranges from -0.516 to 0.022, with no positive significant result; headroom versus NN-K minus Surrogate is negative in all 32 configurations. The NN-K-versus-Random control is mixed, with 5/32 positive significant results. These controls do not support using headroom alone to select among Surrogate, NN-K, and Random.

Headroom remains a valid retrospective description of how far FixedTopK lies from the best candidate observed in the finite pool. It requires labeled historical data and evaluation of the pool, and it does not identify which alternative policy will recover that room. Evidence about policy quality therefore comes from the direct mean-CER, paired-test, and oracle-free selector comparisons reported in the surrounding sections.

6.6 Nearest-Neighbor Transfer Baseline

We also evaluate NN-K as a retrieval-based page-conditioned baseline on the Vietnamese raw, contour-suppressed, and SROIE settings. NN-K substantially outperforms Random K in all evaluated recognizer-by-K comparisons, confirming that OCR-free page features contain useful information for transferring preprocessing choices.

Table 4 Policy-agnostic headroom controls over 32 dataset/setting/recognizer/budget configurations. Coupled outcomes share FixedTopK with headroom; non-shared outcomes are direct policy-to-policy gains. Positive sig. counts positive Spearman correlations with $p < 0.05$; Q4>Q1 counts configurations whose highest-headroom quartile has a larger mean outcome than the lowest.

Analysis	Outcome	Spearman r range	Positive sig.	Q4>Q1
Coupled	FixedTopK − Surrogate	[0.324, 0.638]	32/32	32/32
Coupled	FixedTopK − Random	[0.359, 0.843]	32/32	32/32
Coupled	FixedTopK − NN-K	[0.395, 0.759]	32/32	32/32
Non-shared	Random − Surrogate	[−0.516, 0.022]	0/32	1/32
Non-shared	Random − NN-K	[−0.286, 0.191]	5/32	11/32
Non-shared	NN-K − Surrogate	[−0.370, −0.066]	0/32	0/32

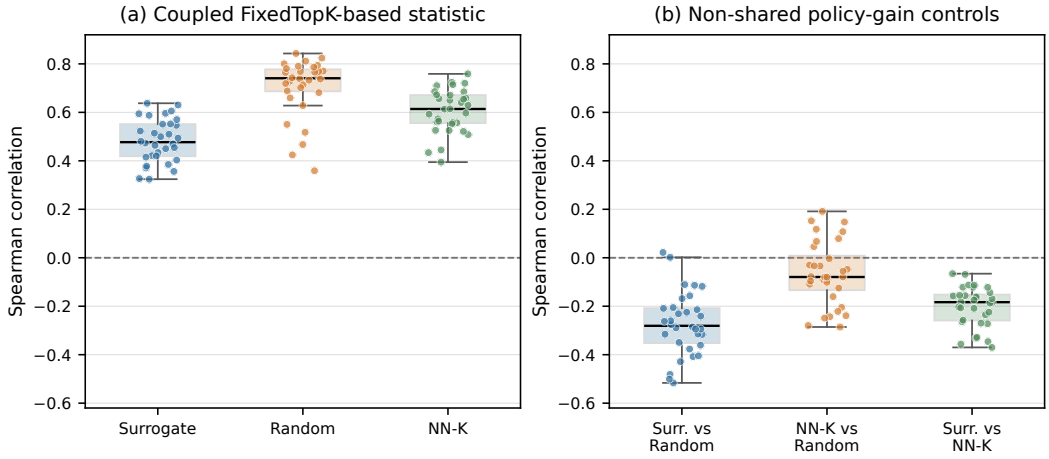


Fig. 2 Spearman correlations across all 32 dataset/setting/recognizer/budget configurations. Each point is one configuration and the dashed line marks zero. (a) Headroom versus FixedTopK-relative advantage is positive for Surrogate, Random, and NN-K, consistent with the shared FixedTopK term and oracle constraint. (b) Direct policy-to-policy gains remove the shared term and show no corresponding positive pattern for learned ranking. Boxes show the median and interquartile range.

Against the learned surrogate, however, there is no uniformly dominant page-conditioned policy. The surrogate is stronger for raw EasyOCR and for SROIE Tesseract mainly at smaller budgets; the two methods are broadly comparable for Tesseract in the Vietnamese settings; and NN-K is competitive or better for VietOCR and for SROIE EasyOCR at the smallest budgets.

The absolute NN-K-versus-surrogate differences are generally small in the Vietnamese settings, especially after contour suppression, where baseline CER is already low. Full paired NN-K comparisons and p-values are provided in the public result tables. The main conclusion is therefore not that surrogate regression dominates retrieval-based transfer, but that both non-random page-conditioned policies can improve over random shortlists and that their ordering depends on recognizer, domain, preprocessing condition, and budget.

6.7 VietOCR as the Strongest Recognizer

For VietOCR, the gain over random at $K = 3$ is smaller in absolute terms because the baseline CER is already low. However, Surrogate $K = 3$ still improves over both NoPre and FixedBest in best-of-K evaluation, showing that adaptive shortlist ranking remains useful in the low-budget $K =$

3 setting even for the strongest recognizer. In the raw setting, VietOCR improves from NoPre 4.70% and FixedBest 4.63% to Surrogate $K = 3$ 4.23%, compared with Random $K = 3$ 4.53%. In the contour-suppressed setting, VietOCR improves from NoPre 3.15% and FixedBest 3.19% to Surrogate $K = 3$ 2.89%, compared with Random $K = 3$ 3.08%.

6.8 Contour Suppression as a Complementary Stage

Contour suppression and surrogate ranking address different failure modes and are complementary rather than alternative preprocessing strategies. The contour rule suppresses large rectangular non-text regions before detection, whereas the surrogate ranks photometric preprocessing settings within the fixed candidate pool. Contour suppression alone improves NoPre CER from 4.70% to 3.15% for VietOCR, 9.82% to 7.69% for Tesseract, and 15.44% to 13.25% for EasyOCR. The combined contour + surrogate setting further reduces best-of-K CER relative to contour NoPre and contour FixedBest: at $K = 15$, it reaches 2.50%, 5.49%, and 11.37% CER for VietOCR, Tesseract, and EasyOCR, respectively. A schematic illustration of the contour-suppression operation is provided in Online Resource 1.

6.9 Feature Ablation

The feature ablation evaluates which feature groups improve surrogate quality and whether the surrogate-vs-random conclusion depends on any single feature group. The document-aware features are the five page-structure descriptors introduced in Section 4: text density, skew angle, connected-component count, aspect ratio, and estimated line count. The four-model ablation rerun confirms that these features matter most for Tesseract: removing them reduces Tesseract global Spearman and R^2 from 0.674 and 0.614 to 0.512 and 0.229 in the raw setting, and from 0.586 and 0.576 to 0.508 and 0.369 in the contour-suppressed setting. EasyOCR also degrades without the document-aware features, but more moderately, while VietOCR changes are small and mixed.

Interaction terms have limited quantitative impact. Removing them changes global Spearman by only about ± 0.015 and $K = 3$ CER by at most about 0.13 percentage points across the raw and contour-suppressed settings. Across every ablation condition, all 12 recognizer-by-K surrogate-vs-random comparisons remained significant in both reruns. Thus, the ablations affect surrogate quality metrics more than they affect the main paired-test conclusion that surrogate-ranked shortlists outperform random shortlists.

6.10 SROIE Evaluation: Surrogate Outperforms FixedTopK

The SROIE evaluation applies the same full-page shortlist-ranking protocol to a different document domain and English OCR configuration. Unlike the Vietnamese settings, the surrogate directly outperforms FixedTopK across all evaluated SROIE recognizer-budget comparisons. SROIE also has greater aggregate FixedTopK-to-oracle headroom, but Section 6.5 shows that headroom alone does not explain which policy will perform best.

For compactness, Table 5 shows $K = 3$ and $K = 15$; the $K = 5$ and $K = 10$ results follow the same qualitative pattern and are provided in Online Resource 1 as well as the public result tables. All eight surrogate-vs-random comparisons are significant, and the surrogate outperforms FixedTopK in all eight SROIE comparisons under two-sided tests. NN-K also improves over FixedTopK and Random K, while the surrogate-versus-NN-K ordering remains recognizer- and budget-dependent: the surrogate is stronger for Tesseract at the two smallest budgets, whereas NN-K is stronger for EasyOCR at the two smallest budgets. The absolute SROIE CER values should not be read as receipt key-information-extraction or cropped-field recognition benchmarks, because the experiment uses full-page OCR output, English OCR configurations, and strict page-level CER without ground-truth text boxes.

6.11 Oracle-Free Top-3 Output Selection

The preceding best-of-K results measure shortlist quality using the best observed CER inside

Table 5 SROIE receipt-setting results. CER values are percentages.

Recognizer	R^2	Global Spearman	Per-image Spearman	NoPre	FixedBest	SampOracle	Surr K=3	NN K=3	FixedTopK K=3	Rand K=3	Surr K=15	NN K=15	FixedTopK K=15	Rand K=15
Tesseract	0.480	0.674	0.588	50.52	46.80	40.17	43.23	43.68	44.27	46.54	40.99	41.03	41.44	42.22
EasyOCR	0.609	0.672	0.664	40.85	40.06	37.88	39.08	38.81	39.22	40.81	38.17	38.14	38.28	38.59

each shortlist. To evaluate an oracle-free operating point, Table 6 reports the nested top-3 selector described in Section 4. This selector uses only the surrogate-ranked top-3 OCR outputs and derived prediction, text-quality, and agreement features at inference time. It is compared with Top1Pred, the agreement-only baseline, FixedBest, and NoPre. All values in the table are one-output CERs rather than best-of-K oracle values.

The selector improves over Top1Pred in mean CER for every row in Table 6. Under two-sided paired tests, the improvement over Top1Pred is significant for all rows except contour-suppressed VietOCR and contour-suppressed EasyOCR. Compared with FixedBest, the selector is significantly better for raw VietOCR and Tesseract, contour-suppressed Tesseract and EasyOCR, and both SROIE recognizers. The raw EasyOCR and contour-suppressed VietOCR differences are lower in mean CER but do not reach $p < 0.05$ against FixedBest. The largest effect appears on SROIE Tesseract, where the selector reduces CER from 46.80% for FixedBest and 46.21% for Top1Pred to 44.29%. These results show that the surrogate top-3 shortlist can support an oracle-free output-selection step, although the gain remains setting-dependent and does not close the full gap to the best-of-3 oracle.

6.12 Feature-Matched Selector Sensitivity Analysis

The main selector uses continuous surrogate predictions and prediction margins, whereas NN-K and FixedTopK supply only a candidate order. This difference could favor the surrogate selector. We therefore repeated the nested selector after replacing the surrogate prediction features with ranks 1, 2, and 3, while preserving the exact top-3 candidates and OCR outputs. The 14 resulting paired comparisons use a separate Holm correction across the feature-matched sensitivity analysis and are reported in detail in Online Resource 1.

Against NN-K, the feature-matched surrogate selector remained significantly better after correction for raw VietOCR (difference −0.21 percentage points), raw EasyOCR (−0.34 points), and contour-suppressed EasyOCR (−0.43 points). The other three Vietnamese NN-K comparisons were not significant; contour-suppressed Tesseract was effectively tied, with a difference of +0.01 points. Against FixedTopK, none of the six Vietnamese feature-matched comparisons was significant after correction. On SROIE, the feature-matched surrogate selector remained better than FixedTopK for Tesseract (44.27% versus 45.13%, difference −0.86 points, 95% CI [−1.20, −0.52]), while the EasyOCR difference was negligible and not significant. A corresponding SROIE NN-K selector comparison is not reported because the stored NN-K shortlist cannot be reconstructed exactly for three EasyOCR pages. These results show that the strongest selector findings are not explained only by continuous surrogate scores and prediction margins, while the comparison with a strong static shortlist remains setting-dependent.

7 Discussion

7.1 Why Ranking Works Despite Moderate R-squared

Moderate R^2 does not preclude useful shortlist construction: the surrogate's task is to place strong candidates near the top of the ranked pool, not to estimate every CER perfectly. The nested results bear this out—moderate-to-strong per-image Spearman together with significantly better-than-random shortlists in all raw and contour-suppressed recognizer-by-K comparisons. For a fixed candidate pool under a limited OCR budget, useful within-page relative ordering is more important than highly accurate absolute CER prediction.

7.2 Interpreting Policy Comparisons and Available Headroom

The direct comparisons show why policy choice must be evaluated rather than inferred from one summary statistic. FixedTopK transfers well on the Vietnamese administrative data, particularly after contour suppression, whereas the surrogate consistently improves over FixedTopK on

Table 6 Oracle-free top-3 output-selection results. CER values are percentages. Sel-Fixed is Selector minus FixedBest, so negative values favor the selector. Reported p-values are two-sided paired Wilcoxon tests for Selector versus FixedBest.

Setting	Recognizer	Selector	Top1Pred	Agreement	FixedBest	NoPre	Sel-Fixed	p(Selector vs FixedBest)
Raw	VietOCR	4.52	4.66	4.60	4.63	4.70	-0.11	9.18×10^{-5}
Raw	Tesseract	8.97	9.47	9.11	9.29	9.82	-0.32	0.0126
Raw	EasyOCR	14.98	15.19	15.27	15.19	15.44	-0.22	0.112
Contour	VietOCR	3.19	3.20	3.16	3.19	3.15	-0.00	0.142
Contour	Tesseract	7.04	7.58	7.12	7.28	7.69	-0.23	0.0151
Contour	EasyOCR	12.92	13.01	13.07	13.09	13.25	-0.17	0.0313
SROIE	Tesseract	44.29	46.21	45.68	46.80	50.52	-2.51	4.63×10^{-37}
SROIE	EasyOCR	39.80	40.13	40.03	40.06	40.85	-0.26	0.000563

SROIE. NN-K is also competitive and sometimes better than the surrogate. These results establish a setting-dependent ordering among static, retrieval-based, and learned policies; they do not establish a universal winner.

The Vietnamese corpus comes from one government portal and has more consistent source and layout characteristics than the receipt images in SROIE. This is compatible with the strong transfer of a static shortlist on the Vietnamese data and the larger gains from page-conditioned policies on SROIE. However, the headroom controls prevent a stronger causal interpretation: FixedTopK-to-oracle headroom correlates with every FixedTopK-relative policy advantage, including Random, because the quantities share a term and are constrained by the sampled-pool oracle.

Headroom is therefore useful only as a retrospective measure of available improvement within the sampled pool. If it is uniformly small on a labeled historical collection, no pool-constrained policy can improve much over FixedTopK. If it is large, an alternative policy has room to improve, but direct held-out comparisons are still required to determine whether Surrogate, NN-K, or another policy actually recovers that room. Headroom is not an oracle-free deployment feature or a policy-selection rule.

The contour-suppressed VietOCR results at larger K also illustrate a complementary point about effect size: at $K = 10$ and $K = 15$, the surrogate and random shortlists produce almost identical mean CER because both are already close to the best available candidates. Paired tests may still show a directional advantage, but the mean difference is at most about 0.03 percentage points and is practically negligible.

7.3 Surrogate Ranking and Retrieval-Based Transfer

The NN-K baseline provides an additional perspective on what is learned by the surrogate. NN-K is also page-conditioned, but it transfers candidates from similar training pages instead of learning a parametric CER-ranking function over (i, θ_j) pairs. Its strong performance against Random K confirms that page-level visual features are informative for preprocessing selection. Its comparison with the surrogate is more nuanced: the surrogate is better for raw EasyOCR and SROIE Tesseract mainly at smaller budgets, the two methods are broadly comparable for Tesseract in the Vietnamese settings, and NN-K is competitive or better for VietOCR and for SROIE EasyOCR at the smallest budgets.

This pattern should be interpreted as empirical rather than causal. It suggests that surrogate ranking and nearest-neighbor transfer exploit partly different aspects of the page-feature space. A learned surrogate can improve over simple similarity transfer in settings such as raw EasyOCR and SROIE Tesseract at smaller K, but retrieval-based transfer can be highly competitive when similar training pages provide good candidate choices, as in the VietOCR settings and SROIE EasyOCR at small K. Thus, the main conclusion is not that surrogate regression dominates all page-conditioned heuristics, but that page-conditioned shortlist construction is useful and that the best non-random policy depends on the recognizer, document domain, and preprocessing condition.

The feature-matched selector analysis gives a related robustness check at the output-selection stage. Replacing the surrogate's numerical prediction and margin features by ranks did not remove its significant advantages over NN-K for raw

VietOCR, raw EasyOCR, and contour-suppressed EasyOCR. It also preserved the SROIE Tesseract advantage over FixedTopK. This reduces concern that those results arise only because the surrogate selector has richer numerical prediction features. However, the analysis does not isolate shortlist quality alone: the selector is refit and still uses OCR text, detection, and agreement features. It should therefore be read as a targeted sensitivity analysis rather than proof that one shortlist policy is intrinsically superior in all settings.

The two evaluation levels should therefore be distinguished. At the best-of-K level, the ordering of Surrogate and NN-K is setting-dependent because the metric measures shortlist quality only. In the feature-matched top-3 selector analysis on the Vietnamese data, the surrogate was significantly better than NN-K in three of six comparisons after Holm correction and did not show a significant disadvantage in the other three. This does not make the surrogate a universal winner. It shows that, once three OCR outputs are available, the surrogate shortlist can support an oracle-free selection policy that remains competitive with NN-K in these settings.

7.4 Practical Interpretation of K

The parameter K is an evaluation budget for shortlist quality: Surrogate K, Random K, FixedTopK, and NN-K report the best observed CER within K candidates, not one-call deployment. This makes small budgets most relevant; at $K = 3$, page-conditioned shortlists already beat random selection, while the surrogate's advantage over FixedTopK and NN-K remains setting-dependent. For oracle-free use, Top1Pred is a one-call option, but its relative performance varies across settings. The nested top-3 selector is a stronger three-call deployment check.

8 Limitations

1. Surrogate K, Random K, FixedTopK, and NN-K CERs measure shortlist quality rather than fully deployable oracle-free performance. The nested top-3 selector provides an oracle-free check, but only for $K = 3$, with three OCR calls per page, and with a supervised selector trained from observed CER. Its primary version also uses continuous surrogate predictions and prediction margins. The feature-matched sensitivity analysis removes this numerical advantage, but it is not a pure shortlist-quality test because the selector is refit and still uses OCR-output features.
2. The surrogate is not a universally best non-random policy. FixedTopK is highly competitive on the Vietnamese administrative data, and NN-K is competitive or better in several recognizer-budget settings.
3. External validity remains limited. The Vietnamese evaluation uses 300 first-page images from one public government portal, while SROIE provides a second domain with English OCR settings; results beyond these sources should be verified separately.
4. The method scope is limited: it is a page-level preprocessing selector before a shared PaddleOCR detector, not a new OCR recognizer. The fixed 50-candidate pool is a compact space-filling design over five grayscale controls; extending it to color-aware enhancement, binarization, morphology, geometric correction, or adaptive candidate generation would require a larger or redesigned pool.
5. Prediction and ranking quality vary across recognizers and folds. Tesseract's per-image Spearman is lower than its global Spearman, indicating that cross-page CER variation contributes to the global statistic and that some pages remain difficult to rank reliably.
6. FixedTopK-to-oracle headroom is retrospective and requires labeled full-pool evaluations. Because it shares FixedTopK with FixedTopK-relative advantages, its coupled correlation cannot identify which alternative policy will perform best.

9 Conclusion

This paper presented a budgeted page-level preprocessing selection framework for OCR. With detection held fixed, Surrogate and nearest-neighbor shortlists significantly outperformed random selection in all evaluated comparisons, while FixedTopK proved to be a strong static baseline and no non-random policy was uniformly best. FixedTopK was highly competitive on the Vietnamese administrative data; on SROIE, the surrogate outperformed FixedTopK in all

eight recognizer-budget comparisons. The nested oracle-free top-3 selector had lower mean CER than FixedBest in all eight settings, with six pairwise two-sided comparisons at $p < 0.05$, showing that part of the shortlist gain can be recovered without reference text. A feature-matched sensitivity analysis further showed that the strongest selector results remained after replacing continuous surrogate predictions and margins with ranks.

The headroom control analysis also refines the interpretation of these results. FixedTopK-to-oracle headroom is a policy-agnostic retrospective measure of room remaining in the sampled pool, not evidence that a learned policy will exploit that room. Practical policy choice should therefore be based on direct held-out comparisons, while deployment requires an oracle-free output-selection mechanism such as the nested selector evaluated here. Together, the controlled policy comparison and selector provide a reproducible basis for evaluating budgeted preprocessing selection without claiming a universal ranking policy or an oracle-free headroom diagnostic.

Declarations

Funding. This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Conflict of interest. The authors declare that they have no conflict of interest.

Data availability. The primary document images are derived from public materials downloaded from the Vietnamese Government document portal (https://chinhphu.vn/he-thong-van-ban). The full derived image set and full reference transcriptions are not publicly redistributed because redistribution permission has not been obtained. The public release contains candidate-pool definitions, OCR-free derived page features, candidate metadata, sanitized evaluation outputs, and statistical-test summaries sufficient to reproduce the reported aggregate analyses; the public repository is available at https://github.com/virustous2/budgeted-ocr-preprocessing-selection. The SROIE receipt dataset follows its own benchmark distribution terms and is not redistributed with this manuscript.

Code availability. The code used for preprocessing candidate generation, surrogate evaluation, baseline comparisons, statistical summaries, and the public derived result tables are available in a public repository under the Apache License, Version 2.0, at https://github.com/virustous2/budgeted-ocr-preprocessing-selection. Raw images, full reference transcriptions, OCR text caches, crop images, and trained model pickle files are excluded from the repository.

Ethics approval. Not applicable.

Consent to participate. Not applicable.

Consent for publication. Not applicable.

References

[1] Bergstra J, Bardenet R, Bengio Y, et al (2011) Algorithms for hyper-parameter optimization. In: Advances in Neural Information Processing Systems 24, pp 2546–2554

[2] Bui CT, Nguyen TH, Nguyen TD, et al (2025) H-ocr: A hybrid ocr approach to vietnamese scanned administrative text recognition. In: 2025 17th International Conference on Knowledge and System Engineering (KSE), pp 1–6, https://doi.org/10.1109/KSE68178.2025.11309655

[3] Cawley GC, Talbot NLC (2010) On overfitting in model selection and subsequent selection bias in performance evaluation. Journal of Machine Learning Research 11(70):2079–2107

[4] Du Y, Li C, Guo R, et al (2020) Pp-ocr: A practical ultra lightweight ocr system. CoRR abs/2009.09941. https://doi.org/10.48550/arXiv.2009.09941, arXiv:2009.09941 [cs.CV]

[5] Garg R, Chaudhury S (2016) Automatic selection of parameters for document image enhancement using image quality assessment. In: 2016 12th IAPR Workshop on Document Analysis Systems (DAS), pp 422–427, https://doi.org/10.1109/DAS.2016.53

[6] Huang Z, Chen K, He J, et al (2019) Icdar2019 competition on scanned receipt ocr and information extraction. In: 2019 International Conference on Document Analysis and Recognition (ICDAR), pp 1516–1520, https:

//doi.org/10.1109/ICDAR.2019.00244

[7] Hutter F, Hoos HH, Leyton-Brown K (2011) Sequential model-based optimization for general algorithm configuration. In: Learning and Intelligent Optimization, Lecture Notes in Computer Science, vol 6683. Springer, pp 507–523, https://doi.org/10.1007/978-3-642-25566-3_40

[8] Jaided AI (n.d.) Easyocr. https://github.com/JaidedAI/EasyOCR, software repository; accessed 2026-04-23

[9] Le A, Lam T, Nguyen D (2025) A survey on vietnamese document analysis and recognition: Challenges and future directions. CoRR abs/2506.05061. https://doi.org/10.48550/arXiv.2506.05061, arXiv:2506.05061 [cs.CV]

[10] Li C, Liu W, Guo R, et al (2022) Pp-ocrv3: More attempts for the improvement of ultra lightweight ocr system. CoRR abs/2206.03001. https://doi.org/10.48550/arXiv.2206.03001, arXiv:2206.03001 [cs.CV]

[11] Loeppky JL, Sacks J, Welch WJ (2009) Choosing the sample size of a computer experiment: A practical guide. Technometrics 51(4):366–376. https://doi.org/10.1198/TECH.2009.08040

[12] McKay MD, Beckman RJ, Conover WJ (1979) A comparison of three methods for selecting values of input variables in the analysis of output from a computer code. Technometrics 21(2):239–245. https://doi.org/10.1080/00401706.1979.10489755

[13] Nguyen N, Nguyen T, Tran V, et al (2021) Dictionary-guided scene text recognition. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp 7383–7392, URL https://openaccess.thecvf.com/content/CVPR2021/html/Nguyen_Dictionary-Guided_Scene_Text_Recognition_CVPR_2021_paper.html

[14] Nguyen SV, Nguyen DA, Pham LSQ (2021) Digitalization of administrative documents: A digital transformation step in practice. In: 2021 8th NAFOSTED Conference on Information and Computer Science (NICS), pp 519–524, https://doi.org/10.1109/NICS54270.2021.9701547

[15] Otsu N (1979) A threshold selection method from gray-level histograms. IEEE Transactions on Systems, Man, and Cybernetics 9(1):62–66. https://doi.org/10.1109/TSMC.1979.4310076

[16] Parvandeh S, Yeh HW, Paulus MP, et al (2020) Consensus features nested cross-validation. Bioinformatics 36(10):3093–3098. https://doi.org/10.1093/bioinformatics/btaa046

[17] Phan Q (n.d.) Vietocr. https://github.com/pbcquoc/vietocr, software repository; accessed 2026-04-23

[18] Pratikakis I, Gatos B, Ntirogiannis K (2011) Icdar 2011 document image binarization contest (dibco 2011). In: 2011 International Conference on Document Analysis and Recognition, pp 1506–1510, https://doi.org/10.1109/ICDAR.2011.299

[19] Sauvola J, Pietikäinen M (2000) Adaptive document image binarization. Pattern Recognition 33(2):225–236. https://doi.org/10.1016/S0031-3203(99)00055-2

[20] Singh P, Vats E, Hast A (2018) Learning surrogate models of document image quality metrics for automated document image processing. In: 2018 13th IAPR International Workshop on Document Analysis Systems (DAS), pp 67–72, https://doi.org/10.1109/DAS.2018.14

[21] Smith R (2007) An overview of the tesseract ocr engine. In: Ninth International Conference on Document Analysis and Recognition (ICDAR 2007), pp 629–633, https://doi.org/10.1109/ICDAR.2007.4376991

[22] Snoek J, Larochelle H, Adams RP (2012) Practical bayesian optimization of machine learning algorithms. In: Advances in Neural Information Processing Systems 25, pp

2951–2959

[23] Souibgui MA, Kessentini Y (2022) De-gan: A conditional generative adversarial network for document enhancement. IEEE Transactions on Pattern Analysis and Machine Intelligence 44(3):1180–1191. https://doi.org/10.1109/TPAMI.2020.3022406

[24] Souibgui MA, Biswas S, Jemni SK, et al (2022) Docentr: An end-to-end document image enhancement transformer. In: 2022 26th International Conference on Pattern Recognition (ICPR), pp 1699–1705, https://doi.org/10.1109/ICPR56361.2022.9956101

[25] Tomasi C, Manduchi R (1998) Bilateral filtering for gray and color images. In: Sixth International Conference on Computer Vision, pp 839–846, https://doi.org/10.1109/ICCV.1998.710815

[26] Varma S, Simon R (2006) Bias in error estimation when using cross-validation for model selection. BMC Bioinformatics 7:91. https://doi.org/10.1186/1471-2105-7-91

[27] Vats E, Hast A, Singh P (2017) Automatic document image binarization using bayesian optimization. In: Proceedings of the 4th International Workshop on Historical Document Imaging and Processing. ACM, pp 89–94, https://doi.org/10.1145/3151509.3151520

[28] Vu XS, Bui QA, Nguyen NV, et al (2021) Mc-ocr challenge: Mobile-captured image document recognition for vietnamese receipts. In: 2021 RIVF International Conference on Computing and Communication Technologies (RIVF), pp 1–6, https://doi.org/10.1109/RIVF51545.2021.9642077

[29] Zuiderveld K (1994) Contrast limited adaptive histogram equalization. In: Heckbert PS (ed) Graphics Gems IV. Academic Press, p 474–485, https://doi.org/10.1016/B978-0-12-336156-1.50061-6